

Stage 11: Shopping Cart & Saved-For-Later Engine Specification

Executive Overview

Stage 11 delivers the unified multi-vendor shopping cart and saved-for-later subsystem for the Alight International Multi-Vendor Marketplace. It supports seamless guest cart browsing, automatic cart merging upon customer authentication, real-time multi-vendor price & tax calculations, stock availability validation, and side-by-side saved-for-later lists.

Key Features & Multi-Vendor Cart Architecture

1. Unified Guest & Authenticated Cart Management

- Guest Sessions:** Opaque `sessionId` cookie/header tracking enables persistent cart browsing prior to user sign-up or login.
- Cart Merging:** When a guest user logs in, calling `/api/v1/cart/merge` merges the guest cart items with the existing customer cart (summing duplicate SKU quantities and preserving saved-for-later states).

2. Multi-Vendor Grouping & Cart Splitting

- Seller-Level Grouping:** Cart items are automatically clustered by `vendor_id` into distinct `VendorCartGroupDto` objects with vendor store name, store slug, subtotal, discount, tax, and shipping estimation.
- Individual Vendor Free Shipping Rules:** Calculates seller shipping thresholds (e.g. Free shipping on orders over ₹10,000 per seller).

3. Saved-For-Later Engine

- Customers can move items out of the active checkout cart into a `savedForLaterItems` list without deleting them.
- Items in the saved-for-later section are excluded from active subtotal, tax, and grand total calculations.
- Price Shift Detection:** Records `priceAtAddition` to alert buyers if an item's price changed while saved.
- One-click restore (`/api/v1/cart/items/{id}/move-to-cart`) revalidates live stock before returning the item to the active cart.

4. Real-Time Pricing & Stock Integrity

- Validates quantity against `warehouse_stock` available levels on add, update, and restore.
- Calculates dynamic promotional discounts via `PricingService` and applicable GST/VAT via `TaxCalculationService`.

Schema Changes (`v37__cart_and_saved_for_later_engine.sql`)

1. `cart_items` Table Extensions: - `is_saved_for_later BOOLEAN NOT NULL DEFAULT FALSE` - `price_at_addition NUMERIC(12, 2)`
2. Indexes Created: - `idx_cart_items_saved` on `cart_items(cart_id, is_saved_for_later)` - `idx_carts_user_session` on `carts(user_id, session_id)`

API Surface (`CartController.java`)

Method	Endpoint	Description	Access
GET	<code>/api/v1/cart</code>	Get cart with active & saved items	Public / User
POST	<code>/api/v1/cart/items</code>	Add product SKU variant to cart	Public / User
PUT	<code>/api/v1/cart/items/{id}</code>	Update quantity of cart item	Public / User
DELETE	<code>/api/v1/cart/items/{id}</code>	Remove item from cart	Public / User
POST	<code>/api/v1/cart/items/{id}/save-for-later</code>	Move item to saved-for-later	Public / User
POST	<code>/api/v1/cart/items/{id}/move-to-cart</code>	Move saved item back to cart	Public / User
DELETE	<code>/api/v1/cart</code>	Clear entire active cart	Public / User
POST	<code>/api/v1/cart/merge</code>	Merge guest cart into user account	Authenticated

Integration Tests

- Verified in `CartAndSavedForLaterIntegrationTest.java` .